

Supplementary Material

SILO: Overall Performance Summary of DE, GA,
and Hybrid Optimization for RFIC Design

TABLE I
COMPARISON OF LC OSCILLATOR PERFORMANCE FOR DE, GA, AND HYBRID OPTIMIZATION

Parameter	DE	GA	Hybrid
f_{osc} (GHz)	5.080616	4.995913	5.060124
PD (mW)	9.336126	9.147608	3.809193
$V_{o,pk}$ (V)	1.394506	1.443161	1.485755
PN @ 1 MHz (dBc/Hz)	-119.5082	-119.0940	-116.2680
FoM (dBc/Hz)	-183.9249	-183.4532	-184.5429
Monte Carlo (MC)			
f_{osc} (μ/σ)	5.081 / 0.01764	4.996 / 0.02959	5.06 / 0.01578
PD (μ/σ)	9.358 / 0.6203	9.167 / 0.4684	3.82 / 0.2231
$V_{o,pk}$ (μ/σ)	1.392 / 0.03282	1.443 / 0.01686	1.485 / 0.01936
PN @ 1 MHz (μ/σ)	-119.5 / 0.9019	-115.9 / 0.2939	-116.2 / 0.8634
FoM (μ/σ)	-183.9 / 0.8652	-180.3 / 0.2849	-184.5 / 0.8906
PVT Analysis			
f_{osc} range (GHz)	[4.988819, 5.188114]	[4.777945, 5.244013]	[4.962056, 5.176002]
PD range (mW)	[4.175332, 19.197120]	[4.007449, 18.033020]	[1.635480, 7.711716]
$V_{o,pk}$ range (V)	[0.840750, 1.889377]	[1.056512, 1.874047]	[1.057830, 1.925874]
PN @ 1 MHz range (dBc/Hz)	[-123.3652, -115.9929]	[-120.3017, -111.9844]	[-119.1978, -113.1950]
FoM range (dBc/Hz)	[-187.9008, -180.4753]	[-184.5822, -177.7700]	[-188.7008, -181.6575]
Yield (%)	70.40	95.80	76.60
Runtime (h)	3.69	2.78	2.51
VDD (V)	1.8	1.8	1.8

TABLE II
COMPARISON OF RF-DC RECTIFIER PERFORMANCE FOR DE, GA, AND HYBRID OPTIMIZATION

Parameter	DE	GA	Hybrid
PCE (%)	92.7231	90.6913	94.987
V_o (V)	0.503295	0.500417	0.522454
P_{IN} (dBm)	-28.8589	-28.7609	-28.9783
ΔV_o (mV)	5.20968	3.0957	5.18771
Area (μm^2)	641	841	634
Dynamic Range (dB)	22.1099	22.4218	23.05
Sensitivity (dBm)	-16.97	-16.05	-18.58
Monte Carlo (MC)			
PCE (μ/σ)	92.45 / 1.85	90.19 / 2.10	94.51 / 1.35
V_o (μ/σ)	0.5030 / 0.00255	0.5002 / 0.00186	0.5219 / 0.00311
ΔV_o (μ/σ)	5.171 / 0.196	3.127 / 0.0934	5.230 / 0.220
P_{IN} (μ/σ)	-28.85 / 0.24	-28.41 / 0.30	-28.64 / 0.22
PVT Analysis			
PCE range (%)	[70.60, 95.66]	[70.50, 92.08]	[74.57, 98.04]
V_o range (V)	[0.4719, 0.5238]	[0.4758, 0.5186]	[0.4883, 0.5480]
ΔV_o range (mV)	[4.18, 8.17]	[2.69, 4.61]	[4.33, 8.81]
P_{IN} range (dBm)	[-29.06, -28.10]	[-29.09, -27.98]	[-29.00, -28.25]
Yield (%)	57.70	43.30	69.10
S_{11} (dB)	-23.89	-23.89	-26.08
Runtime (h)	2.68	3.43	3.82
VDD (V)	0.0	0.0	0.0